BioInteract: Interpretable Drug–Target Interaction Prediction via Residue-Level Cross-Attention with Protein-Language and Physicochemical Priors

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ABSTRACT

Accurate prediction of drug–target interactions (DTIs) can support the prioritisation of candidate pairs, yet published models are usually compared through numbers quoted across papers whose protocols differ, and their attribution outputs are rarely accompanied by an explicit statement of where the model may be trusted. We present BioInteract, a DTI framework combining a chemistry-aware molecular graph encoder, an ESM-2/physicochemical protein residue encoder with a configured domain-label channel, and bidirectional cross-attention, and we evaluate it inside a matched-protocol benchmark on the Davis kinase inhibitor dataset in which every model is trained with one shared partition, seed, optimiser, schedule and threshold rule. The comparison is a three-model ladder that changes one component at a time, so that adding the frozen ESM-2 representation and then replacing concatenation with cross-attention can be read separately. Both steps help, but by protocol-dependent amounts: the protein language model adds 0.120 and 0.063 AUPRC under the random and Target-ID-held-out protocols, whereas cross-attention adds 0.067 and 0.201 AUPRC under the same two protocols. A breadth check across two further architecture families places these differences in context: on the random split the five architectures tested fall within a narrow band, so the entity-held-out protocols carry the informative contrasts. We audit chemical and sequence similarity across all partitions, evaluate exact-sequence-grouped cold-start protocols, and report leakage-aware diagnostic controls. On a strict exact-entity-novel BindingDB cohort, whose construction is documented as a complete row-level filtering waterfall for both the curated-articles snapshot and the full BindingDB release, the frozen Davis checkpoint achieved an AUROC of 0.560 and an AUPRC of 0.340; an applicability-domain analysis attributes this limited transfer principally to ligand and target novelty relative to Davis. The attribution analyses generate hypotheses about model-relevant atoms and residues, but, in line with evidence from models trained with explicit non-covalent interaction supervision, unsupervised attention is not treated here as a recovered interaction mechanism. BioInteract therefore contributes an audited, reproducible and domain-of-applicability-aware DTI reference rather than a new architectural principle.

Keywords

Drug–target interaction; interpretable deep learning; cross-attention; graph neural network; protein language model; model attribution

1 Introduction

The emergence of "AI for Science" has increased expectations that computational models should not merely maximise predictive accuracy, but should also expose the evidence and uncertainty needed for scientific hypothesis generation.[1] This expectation is especially important in drug discovery, where identifying drug–target interactions (DTIs) requires reliable prioritisation together with inspectable outputs that can guide, but not replace, mechanistic follow-up.

Experimental profiling of binding affinities through surface plasmon resonance, isothermal titration calorimetry, or kinase activity assays is laborious and prohibitively expensive. The Davis kinase inhibitor dataset[2] alone required systematic measurement of 72 inhibitors against 442 kinases—a campaign spanning only a fraction of the druggable proteome. Computational DTI prediction can ease this bottleneck by prioritising the most promising candidates and thereby reducing cost and attrition in development pipelines.[3,4] Yet the field has largely remained trapped in a *predictive-accuracy race*: successive deep learning architectures—DeepDTA,[5] GraphDTA,[6] MolTrans,[7] DrugBAN[8]—have incrementally improved benchmark metrics while leaving an important modelling question only partly addressed: *which input features drive a given prediction?*

Pairwise DTI modelling is only one of several complementary routes to the same practical decision. Ligand-based affinity and selectivity profiling can prioritise modulators of a receptor family while explicitly delimiting the applicability domain of the resulting models and validating them on external compounds, as illustrated by ALPACA for cannabinoid receptor modulators;[9] unified frameworks such as DTIAM additionally couple interaction prediction with affinity regression and mechanism-of-action assignment.[10] These examples are cited here for context: they indicate that applicability-domain assessment and external validation, rather than benchmark ranking alone, determine whether a predictive model is usable, and the present study adopts that emphasis for a pairwise DTI classifier.

This question lies at the heart of the broader "AI for Science" agenda. A prediction with no interpretable output is, at best, a hypothesis generator. Moving from *prediction* to hypothesis generation therefore benefits from models that expose internal attribution patterns—for example, attention distributed across drug atoms and protein residues and gradient-based atom scores—rather than returning only a scalar probability. These outputs can prioritise follow-up experiments, but they are not direct measurements of physical contacts or pocket geometry.

Classical DTI methods illuminate the fundamental tension between interpretability and accuracy. Structure-based approaches (molecular docking, molecular dynamics) provide structural inference anchored in three-dimensional protein geometry,[11] yet are constrained by the availability of high-quality crystal structures and the steep computational cost of free-energy calculations. Ligand-based methods (QSAR, molecular fingerprints)[12] circumvent structural requirements at the expense of mechanistic transparency. Neither paradigm fully satisfies the twin demands of modern drug discovery.

Deep learning has transformed predictive capabilities in this domain. Pretrained protein language models such as ESM-2[13] encode evolutionary and structural information at residue resolution, while graph neural networks faithfully capture molecular topology. Concurrently, chemical language models have demonstrated that pretrained molecular representations support accurate and interpretable prediction of reactivity and toxicity,[14] and chemistry-enhanced transformer architectures such as Radical-Net recover atom-level reaction mechanisms.[15] Together, these advances underscore the importance of domain-specific pretraining and chemistry-aware inductive biases in molecular AI.[16]

In this study, "prior knowledge" refers specifically to the sequence information encoded by frozen ESM-2 pretraining together with explicit per-residue physicochemical descriptors. It does not denote curated Pfam, InterPro, binding-site, or other protein-domain annotations, and the manuscript title has been narrowed accordingly.

The "AI for Science" paradigm benefits from attribution mechanisms that are part of the prediction pathway rather than appended only after training. Attention visualisation and SHapley Additive exPlanations (SHAP) can nevertheless have faithfulness limitations.[17] A central principle is therefore that, when attention is an integral component of the prediction pathway, the resulting attributions can be examined consistently with model output but still require external validation before mechanistic interpretation.

Scope of the present contribution

The individual components used here are established. Chemistry-aware molecular graph encoders, residue-level protein language-model features, and cross-modal attention have each been explored extensively, and combinations of them have been published for DTI and drug–target affinity (DTA) prediction.[18–20] We therefore do not claim that any single element of BioInteract, or their combination, is methodologically new. Nor do we claim that its attention maps recover binding mechanisms: as discussed in Section 2, prior work using explicit non-covalent interaction supervision indicates that attention learned only from interaction labels need not be biologically meaningful.[21–23]

What this study contributes is an audited reference point for the Davis binary high-affinity classification task, comprising four elements. First, a matched-protocol benchmark (Section 5.1) in which BioInteract is compared against re-implemented baselines trained under a single shared partition, seed, optimiser, schedule, loss and threshold rule. Its primary form is a three-model ladder that changes one component at a time—molecular graph without a protein language model, then with one, then with cross-attention in place of concatenation—so that each difference has a single cause rather than reflecting a difference in training recipe or reporting convention. Second, a leakage audit of the identifier-based splits together with exact-sequence-grouped cold-start protocols and leakage-aware diagnostic controls. Third, a strict external evaluation on BindingDB whose construction is reported as a complete row-level filtering waterfall for both the curated-articles snapshot and the full database release, accompanied by an applicability-domain analysis that quantifies why transfer is limited. Fourth, model-native attribution outputs released with an explicit statement of what they do and do not establish. Several of these analyses reduce, rather than support, the apparent strength of the model; we report them because a reference point is only useful if its boundaries are visible.

BioInteract rests on three architectural components:

1. **Chemistry-aware molecular graph encoding.** Drug molecules are represented as attributed graphs encoding functional-group-relevant properties (H-bond donor/acceptor status, hydrophobicity, partial charges) that are available to the learned molecular representation.

2. **Protein residue encoding with a disclosed domain-label limit.** Protein representations fuse ESM-2 residue embeddings with per-residue physicochemical properties and a configured domain-label channel. No curated per-protein domain-annotation files are present in the released Davis input package; the channel therefore receives the unknown label at every position and is not interpreted as Pfam/InterPro information.

3. **Bidirectional cross-attention attribution module.** A multi-head cross-attention mechanism computes a matrix $\mathbf{M} \in \mathbb{R}^{n_{\text{atoms}} \times n_{\text{residues}}}$ of learned attention weights between drug-atom and protein-residue representations. The matrix is an integral component of the forward pass; it is used here as a model-native attribution output, not as a predicted atom-level structural contact map.

We evaluate BioInteract on the Davis kinase inhibitor dataset[2] under a random pair split and two entity-held-out protocols: Drug-ID-held-out and Target-ID-held-out. We report chemical and sequence-similarity diagnostics for these partitions, because identifier-level exclusion alone does not guarantee scaffold- or sequence-level novelty. We also add exact-sequence-grouped cold-target and exact-sequence-grouped cold-both evaluations. The attribution analyses—cross-attention maps, Grad-CAM scores, attention-concentration summaries, and case studies—are interpreted as model behaviour rather than as independent evidence of molecular recognition mechanisms.

2 Related Work

2.1 Deep Learning for DTI Prediction

Deep learning approaches to DTI prediction have evolved through several generations.[24] *Sequence-based* models encode both drug SMILES and protein amino acid sequences as one-dimensional strings. DeepDTA[5] employs parallel one-dimensional convolutional neural networks (CNNs) for drug and protein encoding; WideDTA[25] extends this framework with additional molecular descriptors. Although computationally efficient, these models discard molecular topology and are limited in their capacity to model long-range intramolecular interactions.

Graph-based models overcome this limitation by representing drugs as attributed molecular graphs. GraphDTA[6] benchmarks a graph convolutional network (GCN), graph attention network (GAT), graph isomorphism network (GIN), and hybrid GAT–GCN encoder while retaining a CNN for proteins. MolTrans[7] introduces mutual interaction learning between molecular substructure representations and protein subsequences.

Attention-based models include architectures that learn local or cross-modal representations for DTI/DTA prediction. TransformerCPI[26] applies transformer attention to drug–protein interaction patterns. DrugBAN[8] introduces a dual bilinear attention network for fine-grained local interaction modelling. AttentionDTA[27] employs multi-head self-attention for both drug and protein representations. Because these models were published under different splits, binarisation rules, thresholds and reporting conventions, values quoted across their papers are not directly comparable. Section 5.1 therefore re-implements representatives of the sequence-CNN, molecular-graph and bilinear-attention families and trains them under one shared protocol.

2.2 Pretrained Protein Language Models

ESM-2,[13] trained on millions of protein sequences via masked language modelling, produces residue-level embeddings that encode evolutionary conservation, secondary structure propensity, and contact information. These pretrained representations substantially outperform hand-crafted features on downstream tasks.[28] In DTI prediction, pretrained representations have already been combined with graph drug encoders and interaction modules. For example, GEFA uses pretrained protein representations in an early-fusion graph architecture,[18] PGraphDTA replaces a CNN target encoder with protein language-model features,[19] and HCAF-DTA combines protein representations with bidirectional cross-attention.[20] BioInteract therefore does not claim to be the first application of a protein language model, a graph encoder, or cross-attention to DTI/DTA prediction. Because the effect of protein language-model pretraining cannot be separated from architectural choices by comparing against models that lack it, the matched-protocol benchmark in Section 5.1 includes three baselines that consume the identical frozen `esm2_t30_150M_UR50D` representation used by BioInteract and differ only in how drug and protein representations are combined.

Recent DTA studies also illustrate the use of residue-level protein graphs, domain-specific representations, or biological-context features alongside evaluation on more than one benchmark, including 3DProtDTA,[29] the domain-specific framework of Liu *et al.*,[30] and InceptionDTA.[31] These studies, while not directly comparable to the present Davis binary classification protocol, provide further context for alternative approaches to the DTA problem.

2.3 Interpretability in DTI Models

Existing interpretability approaches include post-hoc methods (attention-weight visualisation, gradient-based attribution, and SHAP values[32]) and model-native attribution mechanisms. Attention weights do not necessarily equal feature importance.[17] This general caution has a specific and well-documented form in DTI modelling. MONN showed that jointly supervising non-covalent interaction labels derived from structural complexes is needed to obtain pairwise interaction predictions that agree with those complexes;[21] ArkDTA reached the same conclusion by regularising protein–ligand attention towards non-covalent interaction annotations;[22] and BlendNet reported that interaction-level supervision is what allows a complex-structure-free model to produce attributions consistent with known binding behaviour.[23] We do not treat these studies as a separate category of work that lies outside the scope of comparison. We read them as direct evidence about conventional attention-based models, including BioInteract: absent explicit interaction supervision, there is no reason to expect attention weights to identify the residues that actually contact a ligand, and a small number of qualitative visualisations cannot establish that they do. BioInteract accordingly uses its cross-attention matrix as a model-native attribution that contributes to the prediction and can be inspected alongside it, and we make no claim that this attribution recovers atom-level contacts, binding modes, or physical interaction types. Section 5.8 reports the attribution outputs as descriptions of model behaviour, and Section 6 states the structural evaluation that would be required before any mechanistic reading.

2.4 Cold-Start Evaluation

Random-split evaluation permits information leakage through shared drugs or targets between training and test sets, yielding inflated estimates.[33] Cold-start protocols address this by ensuring that test identifiers are absent from training data. We report the original Drug-ID-held-out and Target-ID-held-out protocols alongside the random split, and separately use exact-sequence grouping for stricter within-Davis stress tests.

3 Methods

3.1 Problem Formulation

Given a drug molecule d represented by its SMILES string and a target protein t represented by its amino acid sequence, we formulate DTI prediction as binary classification: predict whether d binds t with high affinity. Following the established protocol for the Davis dataset,[2] continuous K_d values are binarised at 30 nM ($pK_d > 7.52$), classifying pairs with $K_d < 30\,\mathrm{nM}$ as positive interactions.

3.2 Architecture Overview

BioInteract comprises five sequential modules: (1) a graph isomorphism network with edge features (GINE)-based drug encoder producing per-atom representations, (2) a target encoder combining ESM-2 embeddings with physicochemical features and a configured domain-label channel, (3) a bidirectional cross-attention module computing attention matrices between ligand-atom and protein-residue representations, (4) gated attention pooling to aggregate variable-length representations, and (5) a multilayer perceptron (MLP) prediction head. The full model contains 2,442,083 trainable parameters.

3.3 Drug Molecular Graph Encoder

3.3.1 Chemistry-Aware Atom Featurisation

Each atom is represented by a 52-dimensional feature vector encoding:

- **Topological features** (38 dimensions): atom type (one-hot, 20 types), hybridisation state (6), formal charge (6), number of attached hydrogens (6).
- **Structural features** (9 dimensions): degree, aromaticity, ring membership, and ring size membership (3–8).
- **Functional features** (5 dimensions): hydrogen-bond donor and acceptor status, Crippen logP contribution, molar refractivity, and Gasteiger partial charge.

Bond features (16 dimensions) encode bond type, conjugation, ring membership, and stereochemistry.

3.3.2 GINE Message Passing

We employ a 3-layer GINE architecture:[34]

$$\widetilde{\mathbf{e}}_{vu} = \mathbf{W}_e \mathbf{e}_{vu} + \mathbf{b}_e \in \mathbb{R}^{256}, \qquad \mathbf{e}_{vu} \in \mathbb{R}^{16}, \tag{1}$$

$$\mathbf{h}_v^{(\ell+1)} = \mathrm{MLP}^{(\ell)}\left(\mathbf{h}_v^{(\ell)} + \sum_{u \in \mathcal{N}(v)} \mathrm{ReLU}\left(\mathbf{h}_u^{(\ell)} + \widetilde{\mathbf{e}}_{vu}\right)\right) \tag{2}$$

where $\mathbf{h}_v^{(\ell)}$ is the representation of atom v at layer ℓ, $\mathbf{e}_{vu}$ denotes the raw 16-dimensional edge feature vector, and $\widetilde{\mathbf{e}}_{vu}$ is its learned 16-to-256 projection before message passing. The released `GINEConv` instantiation uses the default fixed $\varepsilon = 0$. Each layer incorporates batch normalisation, ReLU activation, dropout ($p = 0.2$), and a residual connection; the hidden dimension is $D = 256$. Global graph pooling is *not* applied at this stage, preserving individual atom representations for the subsequent cross-attention step.

The archived checkpoints use the stated molecular featurisation; public checkpoint reconstruction applies no stochastic graph augmentation.

3.4 Protein Target Encoder

3.4.1 ESM-2 Residue Embeddings

Residue-level representations are extracted from ESM-2 (150 M parameters, `esm2_t30_150M_UR50D`),[13] yielding 640-dimensional embeddings for each amino acid. These embeddings encode evolutionary conservation, secondary structure propensity, and long-range contact information learned from 65 million protein sequences. ESM-2 parameters are frozen throughout training.

3.4.2 Physicochemical Property Encoding

Each residue is augmented with a 4-dimensional physicochemical vector:

$$\mathbf{p}_i = \left[\text{hydrophobicity}_i,\ \text{mol. weight}_i,\ \text{charge}_i,\ \text{polarity}_i\right] \tag{3}$$

Hydrophobicity follows the Kyte–Doolittle scale.[35] These values provide residue-level physicochemical descriptors; by themselves, they do not identify pocket membership, solvent exposure, or interaction partners.

3.4.3 Configured Domain-Label Channel and Availability

The implementation contains a learnable domain-label embedding layer:

$$\mathbf{d}_i = \text{DomainEmbed}(\text{domain\_type}_i) \in \mathbb{R}^{32} \tag{4}$$

The label builder assigns the unknown `NONE` type when a per-protein annotation file is unavailable. The released Davis package contains no such annotation files, so all archived input positions use this unknown label. We therefore do not represent this channel as curated Pfam/InterPro context, as a binding-site label, or as evidence for a domain-specific conclusion.

3.4.4 Feature Fusion

The three feature sources are concatenated and projected through a shared linear transformation:

$$\mathbf{r}_i = \text{Proj}([\mathbf{e}_i \,\|\, \mathbf{p}_i \,\|\, \mathbf{d}_i]) \in \mathbb{R}^D \tag{5}$$

where $\mathbf{e}_i \in \mathbb{R}^{640}$ is the ESM-2 embedding and Proj is a two-layer MLP with layer normalisation and Gaussian error linear unit (GELU) activation. In the released configuration, $\mathbf{d}_i$ is the shared unknown domain-label embedding rather than a curated per-residue annotation. The projected dimension $D = 256$ matches the drug encoder output.

3.5 Cross-Attention Interaction Module

The cross-attention module produces the model-native attribution used in this study. It computes bidirectional attention between ligand-atom and protein-residue representations, producing an explicit atom–residue attention matrix.

3.5.1 Drug-to-Protein Attention

$$\text{Attn}(\mathbf{Q}_d, \mathbf{K}_p, \mathbf{V}_p) = \text{softmax}\left(\frac{\mathbf{Q}_d \mathbf{K}_p^\top}{\sqrt{d_k}}\right)\mathbf{V}_p \tag{6}$$

where $\mathbf{Q}_d = \mathbf{H}_d \mathbf{W}_Q^d$, $\mathbf{K}_p = \mathbf{R}\mathbf{W}_K^p$, $\mathbf{V}_p = \mathbf{R}\mathbf{W}_V^p$, with $H = 8$ heads and head dimension $d_k = D/H = 32$.

3.5.2 Protein-to-Drug Attention

Symmetrically, protein residues attend over drug atoms:

$$\text{Attn}(\mathbf{Q}_p, \mathbf{K}_d, \mathbf{V}_d) = \text{softmax}\left(\frac{\mathbf{Q}_p \mathbf{K}_d^\top}{\sqrt{d_k}}\right)\mathbf{V}_d \tag{7}$$

3.5.3 Attribution Map Extraction

The atom–residue attribution map is obtained by averaging drug-to-protein attention weights across all heads:

$$\mathbf{M} = \frac{1}{H}\sum_{h=1}^{H} \mathrm{Attn}_h(\mathbf{Q}_d, \mathbf{K}_p) \in \mathbb{R}^{n\times L} \tag{8}$$

Because $\mathbf{M}$ is computed within the forward pass rather than post-hoc, it is a model-native attribution matrix. It is not interpreted as a physical contact map or as an assignment of a specific interaction type to an atom pair.

3.6 Gated Pooling and Prediction

Variable-length representations are aggregated via gated attention pooling:

$$\mathbf{z} = \sum_i \alpha_i \cdot g(\mathbf{x}_i), \qquad \alpha_i = \frac{\exp(f(\mathbf{x}_i))}{\sum_j \exp(f(\mathbf{x}_j))} \tag{9}$$

where $f\colon \mathbb{R}^D \to \mathbb{R}$ and $g\colon \mathbb{R}^D \to \mathbb{R}^D$ are learned two-layer MLPs with Tanh activation. The pooled drug and protein representations are concatenated and passed through a two-hidden-layer MLP ($256 \to 128$; ReLU; dropout $p = 0.3$) followed by sigmoid activation to produce the binding probability.

3.7 Matched-Protocol Comparison Models

To compare architectures without importing values measured under other protocols, we re-implemented a set of baselines and trained each of them, together with BioInteract, inside one harness. Every run uses the same Davis records, the same partition for a given protocol, seed 42, weighted binary cross-entropy with 0.05 label smoothing, AdamW (learning rate 10^{-4}, weight decay 10^{-5}), an effective batch size of 64 formed from micro-batches of 16 with gradient accumulation, automatic mixed precision, gradient-norm clipping at 1.0, a five-epoch linear warm-up followed by cosine decay, a maximum of 30 epochs with early stopping at patience 6 on validation AUROC, and a decision threshold selected once on that protocol's validation predictions. Cross-attention models allocate one atom–residue map per pair, so the micro-batch and mixed-precision settings bound accelerator memory; validation and test inference are run in full precision so that reported metrics do not depend on the mixed-precision training path.

The primary comparison is a three-model ladder in which exactly one component changes at each step, so that each difference has a single cause:

- **GraphDTA**[6] — the molecular graph encoder with a character convolutional protein encoder. Molecular graph but no protein language model.

- **ESM2-GraphConcat** — the same GINE atom encoder, with the convolutional protein encoder replaced by BioInteract's ESM-2/physicochemical residue encoder; the two representations are pooled independently and concatenated. It differs from GraphDTA only by the protein language model.

- **BioInteract** — identical encoders to ESM2-GraphConcat, with concatenation replaced by bidirectional cross-attention. It differs from ESM2-GraphConcat only by the interaction module.

The second and third rows share BioInteract's frozen protein representation exactly. The first-to-second step therefore isolates the contribution of ESM-2 pretraining under a fixed training recipe and a fixed drug encoder, and the second-to-third step isolates the contribution of the cross-modal interaction module. Because GraphDTA is a re-implementation from its published architecture description rather than a run of the authors' released code, it should be read as a controlled reference for its architecture family rather than as a reproduction of the specific numbers reported in that paper. Every comparison reported in this manuscript was measured in this study under the protocol described above; no value is carried over from another publication.

Two further architectures—a sequence convolutional model in the DeepDTA family[5] and the low-rank bilinear attention and bilinear pooling introduced by DrugBAN[8] applied to the same two encoders—were trained by the same harness on the random split as a breadth check across model families. They are reported in Supporting Information Table S14 and are discussed in Section 5.1.

3.8 Training Provenance and Evaluation Thresholds

The archived entity-split checkpoints preserve model architecture but do not store a uniform optimizer, scheduler, warmup, or label-smoothing record. The Random and Drug-ID-held-out logs explicitly record weighted binary cross-entropy, label smoothing of 0.05, AdamW, and a final-run header, whereas the checkpoint-matched Target-ID-held-out trace has the older `run_split.py` header and records only loss and validation AUROC. Its checkpoint does not retain optimizer or loss state. We therefore do not retrospectively assert a single training recipe across all three historical checkpoints (Supporting Information, Training-provenance table).

No historical entity-split model is retrained for the current-release metrics. The canonical reconstruction instantiates each of those three released checkpoints from its embedded model configuration, performs full-precision inference, selects an F1 threshold once on the corresponding validation predictions, and applies that frozen threshold to test predictions. This separation prevents test-set threshold selection. These archived checkpoints are the source of the ablation results in Section 5.2 only. All architecture comparisons in Section 5.1 come from the matched-protocol harness described above and are independent of the archived training histories.

In contrast, the two exact-sequence-grouped strict protocols were trained from initialisation on their respective new training partitions. They use seed 42, the same Davis preprocessing and architecture, weighted binary cross-entropy with 0.05 label smoothing, AdamW (learning rate 10^{-4}; weight decay 10^{-5}), batch size 64, gradient-norm clipping at 1.0, a five-epoch linear warm-up followed by cosine decay, and a maximum of 100 epochs. The checkpoint with the highest validation AUROC was selected with patience 15, after which an F1 threshold was selected on that protocol's validation predictions and frozen for its test evaluation. Complete strict-split training provenance, checkpoint paths, SHA-256 hashes, selected epochs, and thresholds are provided in the Supporting Information Training-provenance table.

3.9 Interpretability Analysis

3.9.1 Residue-Level Attention Profiling

For a given drug–target pair, per-residue attention scores are obtained by summing the interaction map over all drug atoms:

$$s_j = \sum_{i=1}^{n} M_{ij}, \quad j = 1, \ldots, L \tag{10}$$

These values are model-attribution scores and are not assigned a biological hotspot label.

3.9.2 Grad-CAM Functional-Group Attribution

Grad-CAM[36] is adapted to the graph-neural-network (GNN) setting:

$$\mathrm{CAM}_v = \mathrm{ReLU}\left(\sum_k w_k \cdot A_v^k\right), \quad w_k = \frac{1}{|\mathscr{V}|}\sum_v \frac{\partial z}{\partial A_v^k} \tag{11}$$

where A_v^k is channel k of the third (final) GINE message-passing layer and z is the pre-sigmoid positive-class logit. Gradients are therefore taken with respect to the logit, not the post-sigmoid probability. Functional groups are detected with predefined SMARTS patterns (aromatic ring, hydroxyl, amino, amide, carboxyl, sulfonyl, halogen, ether, heterocycle N, and carbonyl). For each group, all RDKit substructure matches are merged into a deduplicated atom set and the group score is the mean of the normalised atom-level scores over that set. An atom can match more than one group pattern. These scores indicate sensitivity of the prediction to the learned representation and do not identify chemical bonds or contacts in a three-dimensional complex.

3.9.3 Attention Sparsity

For each drug–target pair, residue scores are normalised by that pair's maximum:

$$\widetilde{s}_j = \frac{s_j}{\max_k s_k}. \tag{12}$$

Pair-level sparsity is then

$$\mathrm{Sparsity} = \frac{|\{j : \widetilde{s}_j < 0.1\}|}{L}. \tag{13}$$

The reported global value uses all 1,506 positive Davis records, rather than a single test partition, evaluated with the archived `checkpoints/best.pt` checkpoint. Each pair is normalised separately; the 1,301,380 valid-residue scores are then concatenated before the global distribution and fraction are computed. Consequently, all original train/validation/test memberships are represented and longer valid target sequences contribute more residue-level observations.

4 Experimental Setup

4.1 Dataset

The Davis kinase inhibitor dataset[2] comprises 68 drugs (kinase inhibitors) and 442 targets (protein kinases) forming 30,056 drug–target pairs. Binding affinities (K_d) are binarised at 30 nM, yielding approximately 5 % positive interactions.

4.2 Pre-specified External BindingDB Evaluation

To test exact-entity transfer without changing the submitted Davis training protocol, we curated the official BindingDB curated-articles snapshot `BindingDB_BindingDB_Articles_202607`.[37] This snapshot is the subset of BindingDB curated in-house from the primary literature, and it contains 93,712 measurement rows rather than the millions of rows in the complete release. Table 4 reports the full row-level accounting for both this snapshot and, as an independent verification requested during review, the complete `BindingDB_All_202608` release. Filters are applied in a fixed order to a shrinking row set, so the removals and the retained count sum exactly to the input total. We retained only finite, exact numerical K_d measurements in nM for valid, single-chain proteins of at most 1,200 residues and RDKit-valid ligands. Inequality-qualified values were excluded. The dominant reduction is a measurement-type filter rather than a novelty filter: the great majority of BindingDB rows report K_i, IC_{50} or EC_{50} and carry no K_d value at all, and are removed by the first step. The Davis-overlap exclusions, by contrast, remove only a small number of rows. Before replicate reconciliation, any external pair was excluded when its canonical isomeric SMILES occurred anywhere in Davis or when its full normalised amino-acid sequence occurred anywhere in Davis. The resulting cohort is therefore double-novel by exact ligand and target-sequence identity, although it is not a claim of scaffold or remote-homology novelty. Measurements were then grouped by canonical ligand–sequence pair. A pair with replicate measurements on opposite sides of the 30 nM boundary was removed (a pair-level conflict); for every remaining pair, the median exact K_d was retained and binarised as $K_d < 30\,\text{nM}$. The complete row-to-pair accounting is reported in Table S13.

The released `best_random` checkpoint was evaluated without retraining, fine-tuning, calibration, or threshold selection on BindingDB. We used the existing random-validation threshold of 0.5959881544 and the same 640-dimensional ESM-2 representation used for Davis. Only newly extracted external protein embeddings that passed dimension and sequence-length checks were supplied to the frozen model. AUROC and AUPRC confidence intervals use 600 pair-stratified bootstrap replicates with seed 42. Curation and frozen-inference provenance, including source and prediction hashes, are reported in Table S13.

4.3 Applicability-Domain Analysis

To interpret the external result, each curated BindingDB entity is placed relative to the Davis training distribution on two axes. On the ligand axis we compute the maximum Tanimoto similarity between the non-chiral radius-2, 1024-bit Morgan fingerprint of the external compound and those of all 68 Davis compounds. On the target axis we align each external sequence against all 442 Davis sequences with the Biopython `Align.PairwiseAligner` in global mode (match $= 1$, mismatch $= 0$, gap opening $= -1$, gap extension $= -0.5$) and retain the maximum identity, defined as exact-residue matches divided by full alignment length. This is the same identity definition used for the within-Davis cold-target audit, so the internal and external analyses are directly comparable. External pairs are then stratified by each axis, and AUROC, AUPRC, positive counts and mean predicted probability are reported per stratum. A pair is treated as jointly inside the Davis domain when both scores are at least 0.30. Calibration is assessed by comparing the frozen-threshold exceedance rate and the predicted-probability distribution with the observed positive prevalence. The same two nearest-neighbour scores are computed live by the interactive web server for each user-submitted pair.

4.4 Data Splitting Protocols and Similarity Audit

- **Random split** (70/10/20): drug–target pairs are assigned randomly; shared entities may appear across splits.
- **Drug-ID-held-out split**: drugs are partitioned such that all interactions of a given compound identifier appear in exactly one split. This prevents drug-identifier overlap but does not impose a scaffold-similarity threshold.
- **Target-ID-held-out split**: targets are partitioned analogously; test target identifiers are absent from training. This initial entity-held-out protocol does not impose a sequence-similarity threshold.

For the Drug-ID-held-out audit, we computed non-chiral radius-2, 1024-bit Morgan fingerprints with RDKit and reported the maximum Tanimoto similarity between each test compound and the training compounds. For the Target-ID-held-out audit, we used RapidFuzz `fuzz.ratio`, divided by 100, where normalised Indel similarity is $1 - d_{\text{Indel}}/(|x| + |y|)$ and substitutions are represented by one deletion plus one insertion. We computed the maximum value between each test sequence and all training sequences and separately identified exact duplicate sequences. The original Target-ID-held-out partition contained six exact-sequence groups crossing the training and test sets. We additionally constructed exact-sequence-grouped cold-target and exact-sequence-grouped cold-both partitions, assigning all target identifiers with an identical amino-acid sequence to the same split. These protocols use the same Davis records, 70/10/20 partition ratios, and seed 42. Grouping identical sequences removes exact duplicates only; it does not cluster homologues, and Section 6 states explicitly that the resulting partitions are not a homology-reduced benchmark.

For the strict cold-both protocol, the 70/10/20 ratios are applied independently to drug identifiers and exact target-sequence groups. Only pairs for which both entities belong to the same train, validation, or test block are retained; cross-block combinations are excluded to prevent either entity from crossing a partition. Consequently, the retained pair counts are 15,190/264/1,144, and the pair-level proportions are not 70/10/20. Of the 30,056 Davis pairs, 13,458 cross-block pairs are not used in this cold-both evaluation.

To quantify residual target relatedness after exact-sequence grouping, we used Biopython `Align.PairwiseAligner` in global mode (match $= 1$, mismatch $= 0$, gap opening $= -1$, gap extension $= -0.5$) to align each exact-sequence-grouped cold-target test sequence against every training sequence. Identity is the number of exact-residue matches divided by full alignment length; the maximum is retained for each test target. We also implemented leakage-aware diagnostic controls on this same exact-sequence-grouped cold-target partition. A drug-only control uses non-chiral radius-2, 1024-bit Morgan fingerprints in a logistic-regression classifier; a drug-promiscuity control uses each drug's positive prevalence estimated only from training targets; and a label-shuffled drug-only control permutes training labels while leaving test labels unchanged. Because every target identifier in this split is unseen, a target-specific degree computed from its full label column would leak test labels; the only target-only control that avoids that leakage assigns the training positive prevalence uniformly. These controls are diagnostic rather than capacity-matched replacements for BioInteract.

4.5 Evaluation Metrics

For binary classification under severe class imbalance, we report AUROC and AUPRC as primary metrics—AUPRC is particularly informative because it measures positive-class precision across all recall levels. For every reported protocol, F1, precision, and recall use a decision threshold selected once on the corresponding validation precision–recall curve; AUROC and AUPRC remain threshold-free.

4.6 Implementation

BioInteract is implemented in Python 3.10.14 using PyTorch 2.4.0+cu121 and PyTorch Geometric 2.6.1.[38] Protein embeddings are extracted using ESM-2[13] (`esm2_t30_150M_UR50D`, 640 dimensions; `fair-esm` 2.0.0). Molecular featurisation is performed with RDKit 2025.03.6.[39] Revision diagnostics additionally use Biopython 1.87 and RapidFuzz 3.14.5. Training employs automatic mixed precision (AMP) for memory efficiency on a single NVIDIA GPU. Final checkpoint evaluation is run under `torch.inference_mode()` in full precision, using each checkpoint's embedded architecture configuration.

5 Results

5.1 Matched-Protocol Comparison with Baseline Methods

Table 1 and Fig. 1 report the three-model ladder of Section 3.7, each row trained by the same harness on the same partitions. All values in the table were produced in this study; none is quoted from another publication. Supporting Information Table S14 places the same ladder alongside two further architecture families evaluated by the same harness on the random split.

Table 1. Matched-protocol comparison on the Davis dataset (binary classification, $K_d < 30\,\text{nM}$). Every model was trained in this study under one shared partition, seed, optimiser, schedule, loss and validation-selected threshold; no value is quoted from another publication. "PLM" indicates whether the model consumes the frozen `esm2_t30_150M_UR50D` residue representation used by BioInteract. Best value in each column is in **bold**. The Drug-ID-held-out columns are shown for completeness but are not used to rank the models: the shared early-stopping patience halted every run on that protocol at a first-epoch checkpoint (Section 5.1). Per-run provenance is given in Supporting Information Table S16.

			Random		Target-ID-held-out		Drug-ID-held-out	
Model	**Interaction paradigm**	**PLM**	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC
GraphDTA[6]	Graph + sequence CNN	No	0.885	0.358	0.856	0.306	0.679	0.105
ESM2-GraphConcat	Graph + PLM, concatenation	Yes	0.907	0.478	0.878	0.369	0.684	0.097
BioInteract (ours)	Graph + PLM, cross-attention	Yes	**0.913**	**0.545**	**0.934**	**0.569**	**0.713**	**0.126**

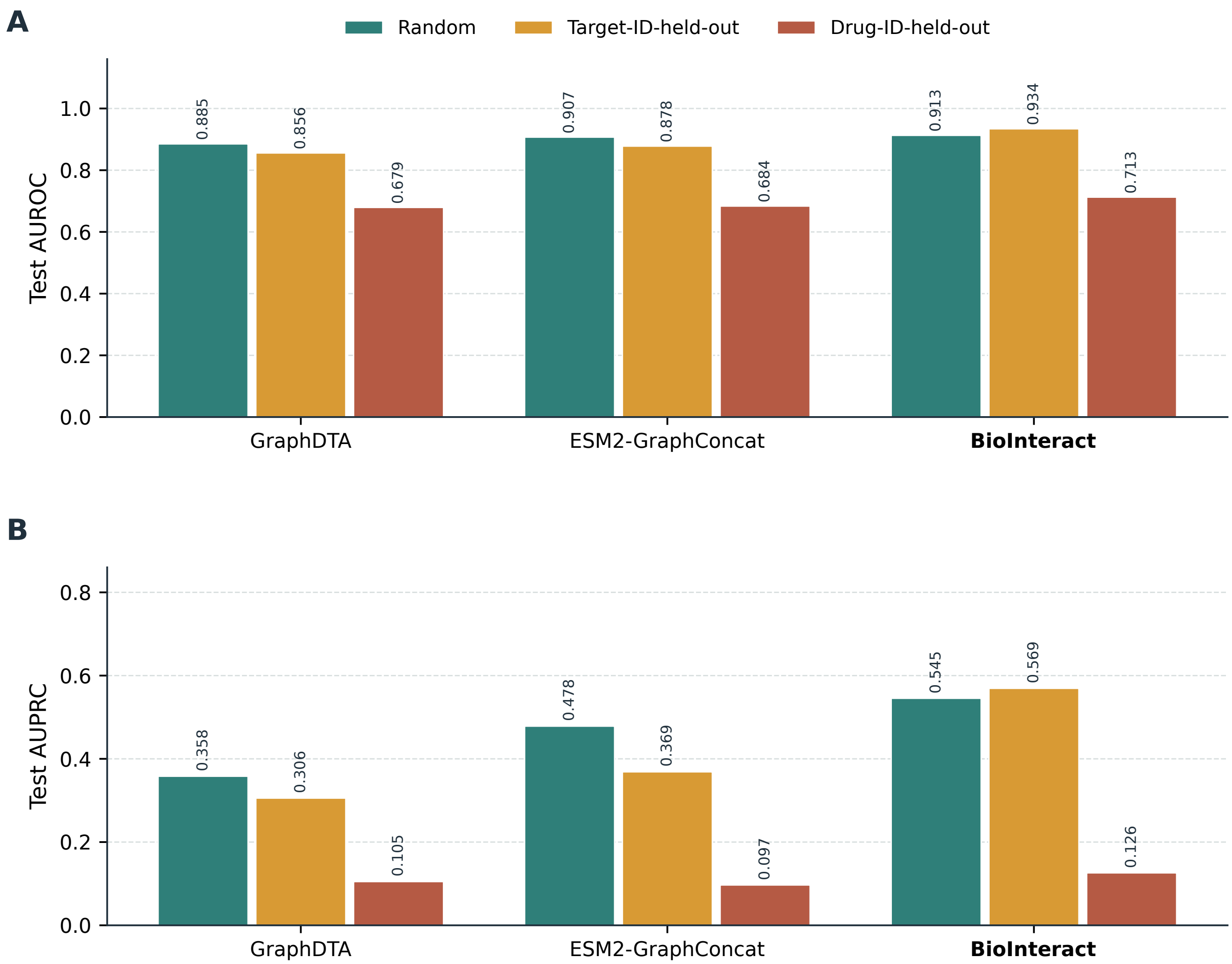

Figure 1. Matched-protocol baseline comparison. Panel A shows test AUROC and panel B shows test AUPRC for each row of the ladder under the random, Target-ID-held-out and Drug-ID-held-out protocols. All values were measured in this study under one shared training protocol. The Drug-ID-held-out bars are shown for completeness only and are not used to rank the models (Section 5.1).

Both steps of the ladder improve on the step below them, but by amounts that depend strongly on the splitting protocol.

One column should be set aside before reading the table. Under the Drug-ID-held-out protocol, validation AUROC dips below its first-epoch value for several epochs in every run, so the shared early-stopping patience of six epochs halted all three models before they recovered and restored a first-epoch checkpoint. That column therefore reflects the harness setting rather than the converged behaviour of the three architectures, and it is not used to rank them; the archived Drug-ID-held-out checkpoint reported in Table S5, trained with a longer patience, reaches AUROC 0.733. The two remaining columns are read below.

Replacing the convolutional protein encoder with the frozen ESM-2 representation, holding the drug encoder and the training recipe fixed, changes AUROC by $+0.022$ and $+0.022$ and AUPRC by $+0.120$ and $+0.063$ under the random and Target-ID-held-out protocols. The contribution of the protein language model is therefore consistent in direction across both usable protocols, and larger on AUPRC than on AUROC.

Replacing concatenation with bidirectional cross-attention, holding both encoders fixed, changes AUROC by $+0.006$ and $+0.056$ and AUPRC by $+0.067$ and $+0.201$ under the same two protocols. The interaction module is thus the smaller of the two effects on the random split and the larger one when target identifiers are held out, where it accounts for most of BioInteract's margin. We read this as the clearest positive result of the matched comparison, subject to the caveat that each configuration was trained once, so differences of a few thousandths are not interpreted.

The Target-ID-held-out column is therefore where the architecture reported here separates most clearly from its own controls: BioInteract reaches AUROC 0.934 and AUPRC 0.569 against 0.878 and 0.369 for the same encoders without cross-attention, and 0.856 and 0.306 without the protein language model as well. Because this is the protocol under which a model would be asked to score a newly characterised target, we regard it, rather than the random split, as the informative column of Table 1. We do not set these values against numbers published for other models: such a comparison requires a shared partitioning, binarisation, threshold rule and model-selection rule, and no controlled mapping between the two exists.

The random split discriminates poorly between architectures, and the breadth check in Supporting Information Table S14 shows why this matters. Under the same harness on that split, the five architectures tested span 0.885 to 0.913 AUROC, a range narrow enough that the ordering within it carries little information. The two additional families were not run on the entity-held-out protocols, so we make no claim about how they behave there. What the random-split breadth check does establish is that the protocol on which DTI models are most often compared is the one on which architectural differences are least visible, and that a favourable random-split number should not be read as evidence for a particular architecture. The results reported here are consistent with that view rather than in tension with it: BioInteract's advantage within the controlled ladder appears where the evaluation is harder, not where it is easiest.

5.2 Ablation Study

We assessed the contribution of the principal architectural components by removing one component at a time from the full model (Fig. 2). These ablation variants come from the archived experiment series rather than from the matched-protocol harness, and their absolute values are therefore not interchangeable with Table 1. They are retained because they cover two components—the domain-label channel and graph augmentation—that the matched-protocol contrasts do not isolate.

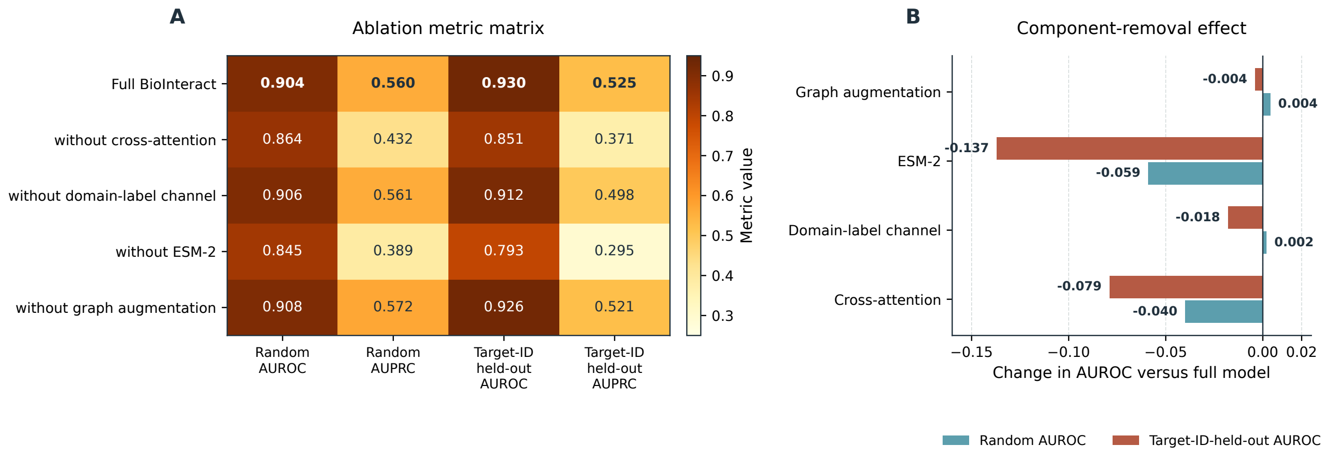

Figure 2. Ablation-study results. Panel A shows the random- and Target-ID-held-out metric matrix for the full model and for each variant with one component removed. Panel B reports the AUROC change after removal of cross-attention, the domain-label channel, ESM-2, or graph augmentation. ESM-2 removal produces the largest AUROC decrease, followed by removal of cross-attention. Values are from the archived experiment series and are not on the same footing as the matched-protocol results in Table 1.

Removing ESM-2 changes AUROC by -0.059 and -0.137 for the random and Target-ID-held-out splits, respectively; removing cross-attention changes AUROC by -0.040 and -0.079. The domain-label and graph-augmentation variants show small mixed changes ($+0.002/-0.018$ and $+0.004/-0.004$, respectively). These ablation results describe the tested Davis configuration and do not establish structural binding mechanisms or out-of-family generalisation.

5.3 Strict Within-Davis Cold-Start Evaluation

The maximum normalised Indel similarity between each Target-ID-held-out test sequence and the original training targets ranges from 0.382 to 1.000 (median 0.579). The audit also found 18 duplicate-sequence groups involving 81 target identifiers in the full dataset and six groups crossing the original train/test boundary. We therefore report the exact-sequence-grouped results separately and limit conclusions to the observed Davis benchmark performance.

Table 2. Strict within-Davis cold-start evaluation after grouping target identifiers with identical amino-acid sequences before partitioning. Both protocols use the original records, seed 42, and the same preprocessing and architecture, but are separately trained from initialisation. For cold-both, 70/10/20 is applied independently to drug IDs and target-sequence groups and only within-block pairs are retained. AUROC and AUPRC are threshold-free test metrics.

Strict protocol	Test pairs	Test positives	AUROC	AUPRC
Exact-sequence-grouped cold-target	5,984	208	0.873	0.383
Exact-sequence-grouped cold-both	1,144	38	0.552	0.038

After grouping identical target sequences, the exact-sequence-grouped cold-target test gives AUROC $= 0.873$ and AUPRC $= 0.383$ (Table 2). Requiring both unseen drug identifiers and exact-sequence-grouped targets is substantially harder (AUROC $= 0.552$, AUPRC $= 0.038$); this test contains 38 positive pairs. These values are reported as an informative within-Davis stress test, not as a claim of out-of-family or structure-level generalisation. Threshold-dependent metrics for these small positive sets are secondary and use a threshold selected only on the corresponding validation set.

The cold-both train, validation, and test blocks retain 15,190, 264, and 1,144 pairs, respectively; 13,458 cross-block pairs are excluded by the double-held-out construction. The validation block contains only 10 positives. At its validation-selected threshold (0.936), the cold-both test yields F1, precision, and recall of 0.000, underscoring that its near-random AUROC and low AUPRC do not support a positive generalisation claim. Full counts and threshold-dependent metrics are reported in Table S7.

The drug-only and drug-promiscuity controls reach AUROC values near 0.764, showing that ligand-side activity or promiscuity signal contributes materially to this within-kinome task (Table 3). BioInteract exceeds these diagnostic controls, but the comparison does not prove that its residue attributions recover physical interactions. The label-shuffled control is near

Table 3. Leakage-aware diagnostic controls on the exact-sequence-grouped cold-target test. All controls use only training-partition labels. They quantify available drug-side signal; they are not capacity-matched substitutes for BioInteract.

Model or control	AUROC	AUPRC
BioInteract (exact-sequence-grouped cold-target)	0.873	0.383
Drug-only Morgan logistic	0.764	0.164
Drug-promiscuity prior	0.764	0.163
Target-only constant prior	0.500	0.035
Label-shuffled drug-only	0.407	0.029

or below the positive-prevalence baseline, confirming that the drug-only signal depends on the observed training labels rather than fingerprint ordering.

For the exact-sequence-grouped cold-target test, the maximum global sequence identity to any training target ranges from 0.231 to 0.867 (median 0.494). Figure 3 stratifies the 88 held-out targets by that maximum identity and reports, for each stratum, BioInteract's AUROC and AUPRC together with the number of test targets, test pairs and positive pairs that support them. Performance is not monotone in identity: the lowest-identity stratum gives the lowest AUROC (0.806), the two middle strata are the strongest (0.908 in both), and the highest-identity stratum falls back to 0.862 with an AUPRC of 0.121 computed from only three positive pairs. The per-stratum positive counts make clear that the two outer strata are the least stable, and the analysis therefore documents the residual relatedness distribution rather than demonstrating robust transfer to homologously distant targets. Numerical values are tabulated in Supporting Information Table S15.

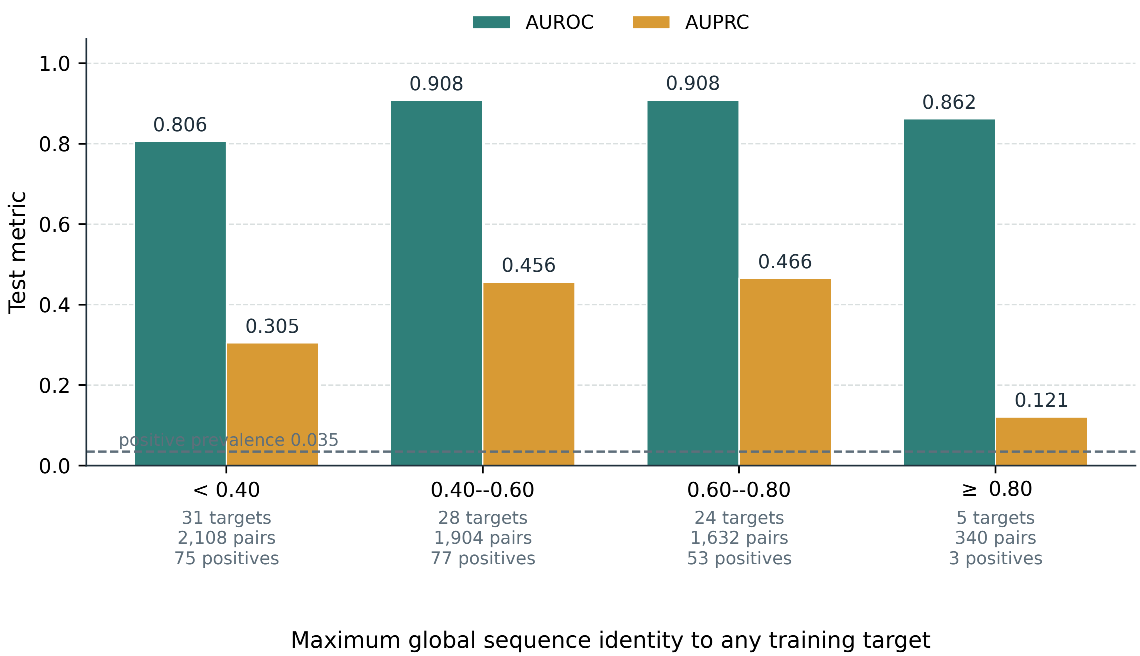

Figure 3. Identity-stratified performance on the exact-sequence-grouped cold-target test. Held-out targets are grouped by their maximum full-length global sequence identity to any training target. Bars give BioInteract's AUROC and AUPRC within each stratum; the annotation under each stratum gives the number of held-out targets, test pairs and positive pairs contributing to it. The dashed line marks the positive prevalence of the full cold-target test. Strata with few positives yield unstable AUPRC estimates.

Under the Drug-ID-held-out split, the maximum test-to-training Morgan Tanimoto similarity ranged from 0.196 to 0.538 (median 0.257). The fingerprint audit is included to clarify that Drug-ID-held-out means unseen compound identifiers, not an externally validated scaffold-hopping or out-of-distribution guarantee.

5.4 Strict External BindingDB Evaluation

5.4.1 Reproducible Construction of the External Cohort

Table 4 gives the complete row-level accounting for the curation described in Section 4.2. The curated-articles snapshot on which the pre-specified evaluation is based contains 93,712 rows, of which 91,062 (97.2 %) report no K_d at all because they record K_i, IC_{50} or EC_{50} instead. Only 2,650 rows carry any K_d value. Every subsequent filter, including both Davis-overlap exclusions, removes at most a few hundred further rows, so the size of the final cohort is determined almost entirely by the requirement of an exact K_d measurement in a snapshot of that size, not by the novelty criteria.

The same pipeline applied to the complete `BindingDB_All_202608` release makes this explicit: from 3,234,499 rows the identical filters retain 52,085 eligible measurements covering 27,079 ligands and 3,472 distinct target sequences, a cohort roughly 24 times larger than the curated-articles cohort. The filter definitions therefore scale as expected, and the modest size of the pre-specified cohort is a property of the snapshot that was fixed before the evaluation rather than of the criteria themselves. Section 5.5 reports the frozen evaluation on that larger cohort. Both waterfalls are reproduced by a released audit script, and the per-snapshot archive hashes are given in Supporting Information Table S17.

Table 4. Row-level filtering waterfall for the external BindingDB cohorts. Filters are applied in the listed order to a shrinking row set, so removals and the retained total sum exactly to the input rows. Supporting Information Table S13 reports the same curation for the curated-articles snapshot with the single-chain test applied first; the ordering changes how rows are attributed between the first three filters but not the eligible total.

	Curated articles, 202607		**Complete release, 202608**	
Filter	Removed	Remaining	Removed	Remaining
Input rows	—	93,712	—	3,234,499
No K_d reported (K_i/IC_{50}/EC_{50} rows)	91,062	2,650	3,104,331	130,168
K_d inequality-qualified or unparsable	160	2,490	47,860	82,308
Target not a single protein chain	27	2,463	2,157	80,151
Ligand SMILES not parsable by RDKit	8	2,455	93	80,058
Non-standard residues or sequence $>$ 1,200 aa	141	2,314	7,644	72,414
Ligand identical to a Davis compound	62	2,252	5,738	66,676
Sequence identical to a Davis target	94	2,158	14,591	52,085
Eligible measurements	2,158		52,085	
Unique eligible ligands / sequences	1,197 / 400		27,079 / 3,472	

We next performed the pre-specified, frozen external evaluation on the curated BindingDB cohort. After all measurement-quality, exact-identity novelty, and replicate-consistency filters, it contains 1,932 pairs from 1,195 ligands and 396 targets: 516 high-affinity and 1,416 low-affinity pairs (Table 5). The model was neither retrained nor recalibrated, and no BindingDB label was used to choose the checkpoint or threshold.

The frozen checkpoint achieved AUROC = 0.560 (95% CI 0.531–0.589) and AUPRC = 0.340 (95% CI 0.314–0.372). At the frozen Davis threshold, F1 was 0.070, precision was 0.731, and recall was 0.037. Thus, although the exact double-novel cohort is a valid external test, it does not support a broad DTI-transfer claim. Instead, it shows limited transfer from this Davis-trained classifier to the broader BindingDB cohort. This external result does not alter the within-Davis checkpoint reconstructions and is not used to interpret attention, residue attribution, binding pockets, or mechanisms.

5.5 Frozen Evaluation on the Complete BindingDB Release

Because the pre-specified cohort is small, we repeated the frozen evaluation on the much larger cohort obtained by applying the identical protocol to the complete `BindingDB_All_202608` release (Table 4). After replicate reconciliation this cohort contains 38,193 double-novel pairs from 26,433 ligands and 3,462 target sequences, of which 13,264 are high-affinity. The same `best_random` checkpoint, the same frozen threshold, and the same freshly extracted `fair-esm` representation were used; no BindingDB label influenced the checkpoint, the threshold, or any preprocessing decision. This evaluation was specified and run after the reviewers' comments, and is reported as a verification of the pre-specified result rather than as a replacement for it.

The frozen checkpoint achieved AUROC = 0.533 (95% CI 0.526–0.539) and AUPRC = 0.383 (95% CI 0.378–0.390) on this cohort, against a positive prevalence of 0.347. At the frozen Davis threshold, F1 was 0.044, precision was 0.513 and recall was 0.023. A cohort twenty times larger therefore confirms the conclusion drawn from the pre-specified evaluation: this Davis-trained classifier retains only weak ranking information outside its training distribution, and its transferred decision threshold is not usable without recalibration.

Table 5. Frozen external evaluation on both BindingDB cohorts. Both use the same `best_random` checkpoint, the same frozen Davis threshold of 0.5959881544, and the same exact-K_d double-novel protocol; they differ only in the source snapshot. Intervals are 600-replicate pair-stratified bootstrap 95% confidence intervals with seed 42.

Cohort	Pairs / positives	AUROC	AUPRC	F1	Recall
Curated articles, 202607 (pre-specified)	1,932 / 516	0.560 [0.531, 0.589]	0.340 [0.314, 0.372]	0.070	0.037
Complete release, 202608 (verification)	38,193 / 13,264	0.533 [0.526, 0.539]	0.383 [0.378, 0.390]	0.044	0.023

5.6 Applicability Domain of the External Cohort

The external cohort was constructed to exclude exact Davis entities, so it is novel by construction; the relevant question is how far novel it is, and whether distance from the Davis training distribution accounts for the loss of performance. Table 6 places every external pair on the two axes defined in Section 4.3.

The cohort is overwhelmingly remote from Davis on both axes. The median external ligand has a maximum Tanimoto similarity of 0.197 to any of the 68 Davis compounds, and 92.6 % of external ligands fall below 0.30. The median external target has a maximum global sequence identity of 0.238 to any of the 442 Davis kinases, and 90.2 % fall below 0.30. This is expected—Davis contains a single protein family—but it means the headline external AUROC is dominated by pairs far outside the training distribution.

Performance tracks that distance closely. In the two lowest ligand-similarity strata, which contain 92.8 % of the pairs, AUROC is 0.509 and 0.504, that is, indistinguishable from chance. In the two strata at or above 0.30 it rises to 0.700 and 0.723. The target axis behaves the same way: AUROC is 0.538 and 0.517 below 0.30 identity and 0.749 and 0.623 above it. Restricted to the 117 pairs that are simultaneously at or above 0.30 on both axes, the frozen checkpoint gives AUROC $= 0.596$ and AUPRC $= 0.719$ against a positive prevalence of 0.641 in that subset. The in-domain signal is therefore real but modest, and the out-of-domain signal is absent.

Table 6. Applicability-domain stratification of the strict external BindingDB cohort. Each pair is placed by the maximum Tanimoto similarity of its ligand to the 68 Davis compounds and by the maximum global sequence identity of its target to the 442 Davis targets. "Entities" counts the distinct external ligands or targets in the stratum. Mean $\hat{p}$ is the mean predicted probability from the frozen checkpoint.

Stratum	Pairs	Positives	Entities	AUROC	AUPRC	Mean $\hat{p}$
Maximum ligand Tanimoto similarity to Davis compounds						
< 0.20	1,076	296	627	0.509	0.305	0.375
0.20–< 0.30	717	144	479	0.504	0.217	0.366
0.30–< 0.40	46	19	36	0.700	0.614	0.420
≥ 0.40	93	57	53	0.723	0.785	0.444
Maximum global sequence identity to Davis targets						
< 0.20	14	1	7	0.538	0.143	0.397
0.20–< 0.30	1,721	415	350	0.517	0.266	0.371
0.30–< 0.40	73	37	15	0.749	0.722	0.409
≥ 0.40	124	63	24	0.623	0.647	0.429
Both axes ≥ 0.30	117	75	—	0.596	0.719	—

A second, independent failure mode is visible in the score distribution rather than in the ranking. The frozen Davis threshold of 0.596 is exceeded by only 1.3 % of external pairs, although 26.7 % of them are positive, which is why the frozen-threshold recall is 0.037 while precision remains 0.731. The predicted probabilities are compressed into a narrow band—median 0.363, 95th percentile 0.437—with a mean of 0.392 for positives against 0.370 for negatives. The operating point transferred from Davis is therefore miscalibrated for this cohort even where the ranking retains information.

Assay and label heterogeneity, by contrast, does not appear to be a major contributor. Only 143 of the 1,932 retained pairs had replicate measurements, and only 15 candidate pairs were discarded because replicate measurements straddled the 30 nM boundary, a pair-level conflict rate of 0.8 %.

The same two nearest-neighbour scores are now computed live by the interactive web server for every submitted ligand–target pair and reported alongside the classifier score with an explicit inside/borderline/outside assessment (Fig. 6), so that a user is told when a query lies in the regime where the external evaluation showed chance-level ranking.

5.7 Training Dynamics

Fig. 4 reports a single checkpoint-matched, contiguous validation-AUROC and training-loss trace for each archived protocol. The duplicated Random trace and two restarted Target-ID fragments were excluded because their epoch counters reset or their peak validation AUROC did not match the released checkpoint. These plots describe observed optimisation paths only; they do not establish absence of overfitting, quantify validation-loss behaviour, or estimate repeated-seed variability.

Figure 4. Training dynamics across splitting protocols. Panel A: validation AUROC over training epochs. Panel B: training loss curves. Each line is one contiguous archived trace whose peak validation AUROC agrees with the saved checkpoint; duplicated and restarted segments are not plotted. No fitted or interpolated trends are shown. The trajectories do not establish absence of overfitting, validation-loss behaviour, or repeated-seed variability.

5.8 Interpretability: Cross-Attention Analysis

All interpretability figures and tables are derived directly from the saved case-level outputs in the project interpretability-results directory, including the report JSON and the corresponding heatmap, profile, and Grad-CAM renderings. Accordingly, we report the model's actual residue-level attention and atom-level Grad-CAM outputs rather than schematic placeholders. These are attribution outputs of the trained model, not independently validated residue contacts or functional-group assignments.

5.8.1 Global Attention Sparsity

Analysis of cross-attention patterns across all 1,506 positive Davis drug–target pairs reveals a global attention sparsity of 99.2 % (Table 7, Fig. 5) under the stated normalisation threshold. Figure 5 is computed directly from the archived 1,301,380 pair-normalised residue scores, not from summary bars; its top 1% and 5% of scores account for 78.9% and 90.4% of their total attribution mass, respectively. This describes concentration of model attribution; it does not demonstrate the size or location of a physical binding interface.

Table 7. Global cross-attention statistics computed across all 1,506 positive Davis pairs using `checkpoints/best.pt`. The values quantify the concentration of the model's pair-normalised residue-attribution scores; they do not provide structural validation of a binding pocket.

Metric	Value
Mean residue attention	0.0044
Median residue attention	0.0003
Top-1% threshold	0.0572
Top-5% threshold	0.0036
Fraction of residues with normalised attribution < 0.1 of the pair-specific maximum	99.2%
Top 1% / top 5% cumulative attribution mass	78.9% / 90.4%
Samples analysed	1,506
Residue-score observations	1,301,380

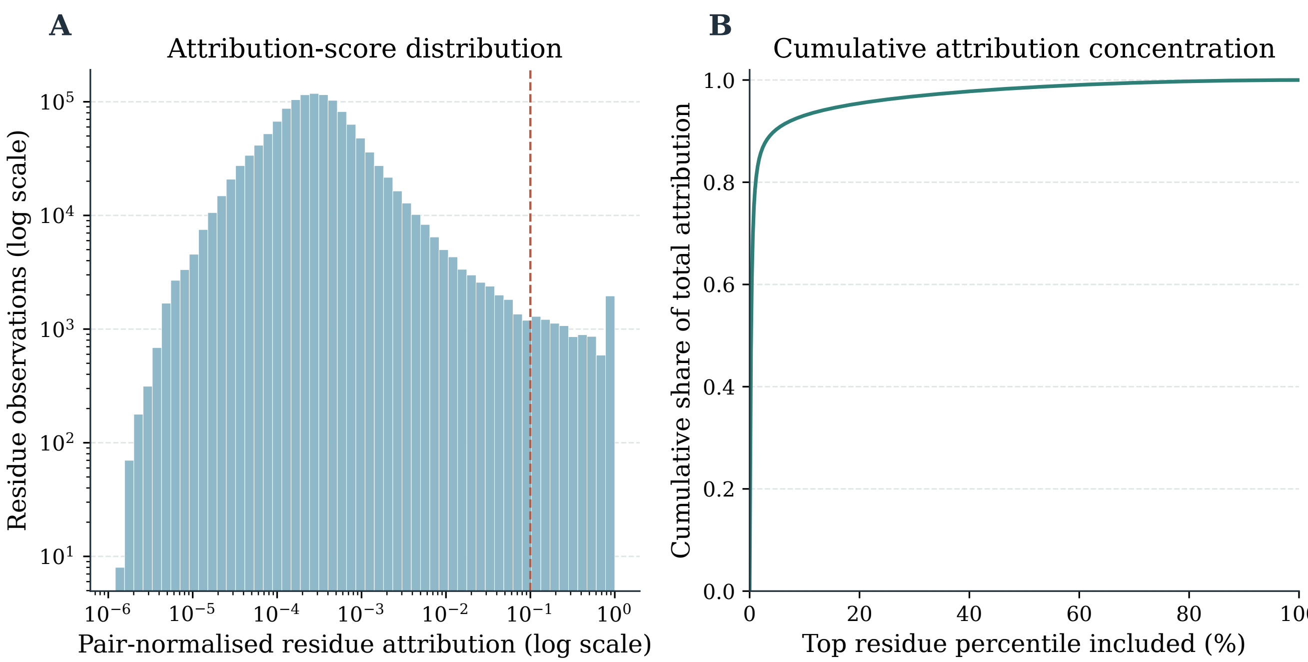

Figure 5. Attention sparsity analysis. (A) Histogram of the archived pair-normalised per-residue attention scores across all 1,506 positive Davis pairs (1,301,380 observations; logarithmic axes); the dashed line denotes the 0.1 sparsity threshold. (B) Cumulative attribution mass after sorting those same scores in descending order. The 99.2 % value is an attribution-concentration statistic, not a claim about a contiguous protein-surface region.

Although softmax attention is dense before normalisation, the observed concentration arises after training on the limited and imbalanced Davis data. It may reflect learned regularities, data composition, or both; it is therefore not presented as independent evidence that the model has recovered molecular recognition principles.

5.8.2 Input-Integrity Audit of Nominal ABL1 Variant Records

This audit tests one falsifiable statement about the benchmark inputs rather than about the model: *if* the Davis records labelled with ABL1 point-mutation names encoded the corresponding sequence substitutions, *then* the amino-acid sequences supplied to the model for those records would differ from the wild-type sequence and from one another. The evaluation criterion is therefore exact: for every nominal ABL1 record we compare the full amino-acid string and its SHA-256 digest in `target_sequences.csv`.

The audit falsifies that statement. Fifteen target records whose nominal names begin with ABL1—including the wild-type and the named F317I, F317L, M351T and E255K records—all carry the same 1,167-residue amino-acid sequence and the same SHA-256 digest (Supporting Information, Table S9). The consequence is a property of the benchmark, not a finding about mutation biology: because variant-specific sequence information is absent from the model inputs, any difference in predicted score between two nominal ABL1 variants can only reflect the paired ligand, and no experiment on these records can measure mutation-specific or resistance-specific model behaviour. The value of the audit is that it defines a boundary on what the Davis benchmark can be used to test, and it is reported for that reason alone; the mutant-comparison analysis that appeared in the original submission is not part of this manuscript.

5.9 Case Study: Functional-Group Attribution

The saved Grad-CAM outputs for the two archived non-mutant Davis cases provide a compact example of atom-level scores aggregated to functional groups (Table 8). The entries are model-native attribution scores rather than measured contacts or interaction types.

Table 8. Model-native functional-group attribution for the two archived non-mutant Davis cases. Scores are normalised atom-level Grad-CAM values aggregated by functional group; they do not identify physical contacts. A dash marks a group with no RDKit substructure match in that ligand.

Functional group	EGFR–D0014 (class score 0.995)	BRAF–D0023 (class score 0.099)
Carbonyl	0.431	—
Amide	0.389	—
Ether	0.136	—
Halogen	0.134	—
Amino	0.090	—
Aromatic ring	0.000	0.059
Heterocycle N	0.000	0.000
Hydroxyl	—	0.000

The two cases differ markedly in classifier score and attribution profile. EGFR–D0014 has a high-affinity-class score of 0.995 and its largest aggregated groups are carbonyl (0.431) and amide (0.389). BRAF–D0023 has a score of 0.099 and only small saved values for aromatic ring (0.059) and hydroxyl (0.000), despite its archived positive affinity of 0.19 nM. We therefore treat BRAF–D0023 as a low-scoring failure-mode example for an experimentally positive pair. This is a descriptive limitation of model behaviour, not an explanation of target binding or an assessment of functional-group chemistry.

6 Discussion

6.1 From Prediction Accuracy to Inspectable Hypotheses

BioInteract is intended to make model behaviour more inspectable than a purely scalar DTI prediction. Its attention matrices and Grad-CAM scores can generate hypotheses about which representations contributed to a prediction. Such hypotheses can subsequently be assessed with crystal structures, mutagenesis, or structure–activity relationship (SAR) experiments; the present study does not replace these measurements.

The matched-protocol comparison sharpens what can and cannot be concluded from the Davis benchmark, and the two conclusions point in opposite directions.

The favourable one is that, within a ladder where a single component changes at a time, both components earn their place, and the cross-modal interaction module earns it where the evaluation is hardest. Cross-attention adds 0.201 AUPRC over concatenation under the Target-ID-held-out protocol against 0.067 under the random split, with both encoders and the entire training recipe held fixed. A comparison run only on the random split would have judged that module nearly worthless.

The qualifying one is that the random split, on which DTI architectures are most often ranked, barely separates them: five architectures trained under the same harness span 0.885 to 0.913 AUROC (Supporting Information Table S14). We therefore treat random-split ordering as weak evidence about architecture, and we do not rest any conclusion on it.

We therefore present BioInteract as one audited point in this design space rather than as a state-of-the-art claim. These Davis results do not show that BioInteract maps a physical binding pocket, assigns an interaction type to an atom pair, ranks a large compound library, or assesses treatment resistance.

6.2 The 99.2% Attention Sparsity

The observed 99.2 % attention sparsity is a property of the pair-normalised, residue-pooled attribution scores under the chosen threshold. Although it is compatible with a model focusing on a small subset of residues, it can also be affected by the kinase-focused and imbalanced training data. It should not be read as evidence that the highlighted residues constitute a true binding site. More generally, work that supervises non-covalent interaction labels directly has shown that concentrated or plausible-looking attention is not by itself evidence of recovered interactions.[21–23] We accordingly report the sparsity statistic as a description of how the trained model distributes attribution mass, and not as support for an interpretability claim.

6.3 Scope of Cold-Start Evaluation

The exact-sequence-grouped cold-target and cold-both evaluations, which provide a strict within-Davis stress test, involve kinase proteins and do not demonstrate performance on unrelated protein families or pocket classes. Equally important, exact-sequence grouping controls duplicate sequences only. Held-out targets retain substantial residual relatedness to the training set—the maximum global identity has a median of 0.494 and reaches 0.867—so these partitions must not be read as a fully homology-reduced cold-target benchmark. A benchmark that clustered targets by kinase-domain similarity, or that adopted a predefined clustered split at a fixed identity cut-off, would provide the stronger control and is a useful direction for future evaluation.

The Drug-ID-held-out result is likewise restricted to the chemical diversity of 68 kinase inhibitors. The observed Morgan-similarity distribution documents the nearest-neighbour relationship between held-out and training molecules, but it does not establish scaffold hopping or out-of-distribution robustness.

6.4 Diagnostic Controls and Remaining Dataset Bias

The exact-sequence-grouped cold-target diagnostic controls demonstrate that a target-independent drug representation retains substantial ranking signal in Davis. The drug-only Morgan and drug-promiscuity controls each achieve AUROC near 0.764, whereas BioInteract reaches 0.873 under the same exact-sequence-grouped partition. This improvement is compatible with the use of target information, but it neither excludes ligand-side activity bias nor establishes that the model has learned a molecular-recognition mechanism. We therefore interpret the control comparison as a boundary on the Davis result, not as causal evidence for any particular encoder or attribution map. The label-shuffled control confirms that the observed drug-only signal depends on the training labels; it is not a test of structural interaction recovery.

6.5 Why External Transfer Is Limited

The external result in Section 5.6 is best read as two separate failures with different remedies.

The first is a domain failure. More than nine-tenths of the external pairs sit below a Tanimoto similarity of 0.30 to any Davis compound and below a sequence identity of 0.30 to any Davis target, and it is exactly those pairs whose AUROC is indistinguishable from chance. Within the small in-domain subset, ranking recovers to AUROC 0.60–0.75 depending on the axis. A model trained on 68 inhibitors and one protein family cannot be expected to rank pairs drawn from the whole of BindingDB, and the stratified result shows that the aggregate AUROC of 0.560 is a mixture of a usable in-domain regime and an uninformative out-of-domain one rather than a uniform level of weak performance. The remedy is broader training data, not a different attribution mechanism.

The second is a calibration failure that is independent of ranking. The decision threshold selected on the Davis validation set is exceeded by 1.3 % of external pairs even though 26.7 % are positive, and the predicted probabilities occupy a narrow band well below that threshold. Any deployment on a new distribution would therefore require the threshold to be re-derived on labelled data from that distribution; reporting a frozen-threshold F1 across distributions, as we do in Table 5, understates ranking quality and overstates the usefulness of the fixed operating point.

Two candidate explanations are not supported by the data. Assay and label heterogeneity is small in this cohort: the pair-level replicate-conflict rate is 0.8 %, and conflicting pairs were removed before evaluation. And the result is not an artefact of cohort size, since the same filters applied to the complete BindingDB release retain 52,085 measurements over 3,472 target sequences (Table 4). We therefore attribute the limited transfer principally to chemical and target novelty relative to Davis, compounded by threshold miscalibration, and we report the applicability-domain scores in the released web server so that this boundary is visible at the point of use.

6.6 Functional-Group Attribution Without Structural Supervision

The Grad-CAM functional-group ranking is qualitatively compared with kinase SAR literature, but it remains a gradient-based attribution. In particular, the analysis does not determine whether a group forms a hydrogen bond, a pi-stacking interaction, a cation–pi interaction, or a van der Waals contact, nor whether an implicated protein residue contributes through its backbone or side chain. These questions require a structural complex or targeted experiments.[40]

6.7 Future Directions

The strict BindingDB evaluation directly indicates limited transfer beyond the Davis benchmark; it therefore cannot support a protein-family-agnostic, broad-pocket, or chemical-space generalisation claim. Evaluation after retraining and validation on a wider, pre-specified multi-family protocol is required to establish whether the limitation arises from the Davis training distribution, task definition, or model. Direct comparison with independent experimental evidence is also needed before attributions can be assigned a structural meaning.

The most informative next step for the attribution outputs is not a further qualitative case study but a supervised comparison. Models trained or regularised with explicit non-covalent interaction labels provide both the benchmark design and the evaluation criterion required.[21–23] A future structural audit should use a non-overlapping benchmark of experimentally resolved complexes, predefine the atom–residue contact criterion, and compare model attributions with the resulting contact labels (for example, from PLIP[41]) before structural meaning is assigned. No overlap analysis between the Davis cases and PDBBind or LP-PDBBind, and no PLIP contact validation, was performed in this study.

The present implementation was audited on one NVIDIA GPU. Although the 2,442,083 trainable parameters do not increase with the number of training pairs, data-processing time, graph batching, ESM-2 embedding extraction and storage, and accelerator memory must be profiled at larger scale. Distributed training was not evaluated, and no claim is made that the current implementation scales linearly to BindingDB-sized data.

Evaluation on curated allosteric and cryptic-site benchmarks, with appropriate structural or dynamic context, is likewise required before BioInteract can be assessed for non-canonical drug–target interactions. Existing work highlights the need for explicit allosteric-site labels and benchmark design,[42,43] and for structural or dynamic treatment of cryptic pockets.[44] CryptoBench provides a dedicated cryptic protein–ligand benchmark that can help define such a future evaluation.[45]

Extending BioInteract from binary interaction classification to continuous binding-affinity regression (pK_d) would provide the finer-grained potency estimates needed for lead optimisation, where accurate ranking of closely related analogues is as important as binary activity prediction.

7 Conclusion

We have presented BioInteract, a DTI-prediction framework that combines chemistry-aware molecular graph encoding, ESM-2/physicochemical residue representations, and bidirectional cross-attention. The model provides prediction scores together with inspectable model-native attention and Grad-CAM attributions.

This revision states the evidence boundary of those outputs explicitly. Under a matched training protocol, both components of the three-model ladder contribute, and the cross-modal interaction module contributes most under the entity-held-out protocols rather than on the random split. Because the random split separates architectures only weakly, we make no claim of general superiority and treat random-split rankings as weakly informative about architecture. The observed attention concentration and functional-group ranking describe model behaviour, and, in view of results obtained with explicit non-covalent interaction supervision, we do not present them as recovered binding contacts or mechanisms. We distinguish Target-ID-held-out performance from exact-sequence-grouped cold-start evaluation after detecting duplicate sequences in the original target partition, while noting that neither is a homology-reduced benchmark. The strict external BindingDB test shows limited transfer of the frozen Davis checkpoint, and the accompanying applicability-domain analysis identifies the conditions under which the released model should and should not be trusted. Broader datasets, clustered splits and direct structural validation remain necessary for generalisation and mechanistic claims.

BioInteract is available as a publicly accessible interactive web interface at `https://huggingface.co/spaces/AI4deeperScience/BioInteract` for the custom-prediction workflow shown in Fig. 6. The repository README documents the command-line workflow for the available attribution outputs.

BioInteract represents a step toward AI-assisted DTI prediction in which model scores are accompanied by transparent attribution outputs and explicit limits on their interpretation. Such outputs are most useful when combined with rigorous split auditing, external validation, and biochemical experiments.

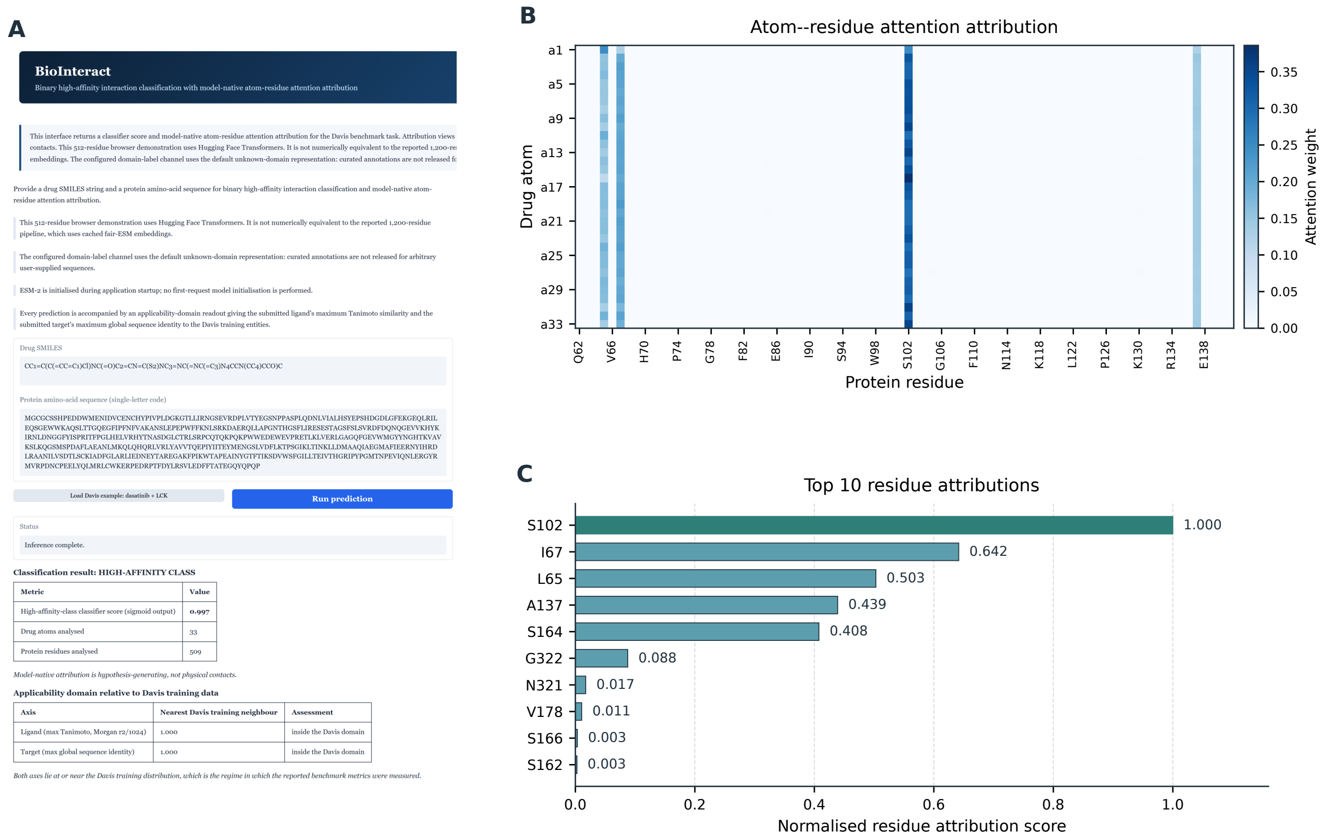

Figure 6. BioInteract interactive web server. Panel A shows the input and result state of the publicly accessible interface at `https://huggingface.co/spaces/AI4deeperScience/BioInteract` for the released Davis example pair D0017 (dasatinib) and T0210 (LCK, 509 residues), including the high-affinity-class classifier score and the applicability-domain readout. For every submitted pair the interface now reports the ligand's maximum Tanimoto similarity and the target's maximum global sequence identity to the Davis training entities, together with an explicit inside/borderline/outside assessment. Panels B and C show the two model-native attribution views returned for the same pair: the atom–residue attention map over the highest-attribution residue window, and the ranked top-ten residue attributions. All three panels are rendered as vector graphics from the released pipeline. These are model-native attribution views, not physical-contact or hotspot labels. Command-line instructions for generating the available attribution outputs are provided in the public repository.

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Supporting Information

The Supporting Information contains: model hyperparameters and feature details (Tables S1–S3); computational environment (Table S4); dataset and split statistics (Tables S5–S7); atom features (Table S8); the Davis ABL1 nominal-variant input audit (Table S9); non-mutant EGFR and BRAF case summaries (Tables S10–S11); complete model parameter counts (Table S12); strict BindingDB external-validation provenance and results (Table S13); a random-split breadth check across model families (Table S14); the numerical identity-stratified cold-target results (Table S15); matched-protocol comparison provenance (Table S16); and the complete BindingDB filtering waterfall for both snapshots (Table S17). This material is available free of charge.

Data Availability

All data used in this study are publicly available. The Davis kinase inhibitor dataset used for model training and evaluation is available from the original publication.[2] The external evaluation uses the official BindingDB curated-articles TSV snapshot `BindingDB_BindingDB_Articles_202607`, obtained from the BindingDB download service;[37] the raw archive remains available from its source, and the release provides the deterministic curation and frozen evaluation scripts plus manifest hashes. The verification waterfall additionally uses the complete `BindingDB_All_202608` release from the same download service. The immutable release containing the released checkpoints, canonical prediction CSVs, provenance hashes, strict-split artifacts, and reproduction scripts is archived at Zenodo, `https://doi.org/10.5281/zenodo.21845772`. The BioInteract source repository is hosted on GitHub. An interactive web server for the custom-prediction workflow is available through the BioInteract Space; the README documents the command-line attribution workflow.

Code Availability

The BioInteract source code, pretrained model weights, configuration files, canonical metric-reconstruction scripts, and figure-generation scripts are publicly available through GitHub and as the immutable Zenodo release, `https://doi.org/10.5281/zenodo.21845772`. The matched-protocol baseline implementations, the BindingDB filtering-waterfall audit, and the applicability-domain analysis added in this revision are included in the same repository and archive. The interactive web application for the custom-prediction workflow is available through the BioInteract Space.

Author Contributions

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Declaration of Competing Interests

The authors declare no competing interests.

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References

1. Schneider, P. *et al.* Rethinking drug design in the artificial intelligence era. *Nat. Rev. Drug Discov.* **19**, 353–364, DOI: 10.1038/s41573-019-0050-3 (2020).
2. Davis, M. I. *et al.* Comprehensive analysis of kinase inhibitor selectivity. *Nat. Biotechnol.* **29**, 1046–1051, DOI: 10.1038/nbt.1990 (2011).
3. Chen, X. *et al.* Drug–target interaction prediction: databases, web servers and computational models. *Briefings Bioinforma.* **17**, 696–712, DOI: 10.1093/bib/bbv066 (2016).
4. Yamanishi, Y., Araki, M., Gutteridge, A., Honda, W. & Kanehisa, M. Prediction of drug–target interaction networks from the integration of chemical and genomic spaces. *Bioinformatics* **24**, i232–i240, DOI: 10.1093/bioinformatics/btn162 (2008).
5. Öztürk, H., Özgür, A. & Ozkirimli, E. DeepDTA: deep drug–target binding affinity prediction. *Bioinformatics* **34**, i821–i829, DOI: 10.1093/bioinformatics/bty593 (2018).
6. Nguyen, T. *et al.* GraphDTA: predicting drug–target binding affinity with graph neural networks. *Bioinformatics* **37**, 1140–1147, DOI: 10.1093/bioinformatics/btaa921 (2021).
7. Huang, K., Xiao, C., Glass, L. M. & Sun, J. MolTrans: Molecular interaction transformer for drug–target interaction prediction. *Bioinformatics* **37**, 830–836, DOI: 10.1093/bioinformatics/btaa880 (2021).
8. Bai, P., Miljkovic, F., John, B. & Lu, H. DrugBAN: a dual bilinear attention network for drug–target interaction prediction. *Nat. Mach. Intell.* **5**, 126–136, DOI: 10.1038/s42256-022-00605-1 (2023).
9. Delre, P. *et al.* ALPACA: a machine learning platform for affinity and selectivity profiling of cannabinoids receptors modulators. *Comput. Biol. Medicine* **164**, 107314, DOI: 10.1016/j.compbiomed.2023.107314 (2023).
10. Lu, Z. *et al.* DTIAM: a unified framework for predicting drug–target interactions, binding affinities and drug mechanisms. *Nat. Commun.* **16**, 2548, DOI: 10.1038/s41467-025-57828-0 (2025).
11. Kitchen, D. B., Decornez, H., Furr, J. R. & Bajorath, J. Docking and scoring in virtual screening for drug discovery: methods and applications. *Nat. Rev. Drug Discov.* **3**, 935–949, DOI: 10.1038/nrd1549 (2004).
12. Cherkasov, A. *et al.* QSAR modeling: where have you been? where are you going to? *J. Medicinal Chem.* **57**, 4977–5010, DOI: 10.1021/jm4004285 (2014).
13. Lin, Z. *et al.* Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **379**, 1123–1130, DOI: 10.1126/science.ade2574 (2023).
14. Huang, Z. *et al.* Large-scale chemical language models for accurate and interpretable prediction of reactivity and toxicity. *Process. Saf. Environ. Prot.* **204**, 108113, DOI: 10.1016/j.psep.2025.108113 (2025).
15. Huang, Z. *et al.* Radical-Net: A chemistry-enhanced Transformer for elementary radical reactions in pollutant chemistry. *J. Hazard. Mater.* **501**, 140631, DOI: 10.1016/j.jhazmat.2025.140631 (2026).
16. Huang, Z. *et al.* Ai-enhanced chemical paradigm: From molecular graphs to accurate prediction and mechanism. *J. Hazard. Mater.* **465**, 133355, DOI: 10.1016/j.jhazmat.2023.133355 (2024).
17. Jain, S. & Wallace, B. C. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 3543–3556, DOI: 10.18653/v1/N19-1357 (2019).
18. Nguyen, T. M., Nguyen, T., Le, T. M. & Tran, T. GEFA: early fusion approach in drug–target affinity prediction. *IEEE/ACM Transactions on Comput. Biol. Bioinforma.* **19**, 718–728, DOI: 10.1109/TCBB.2021.3094217 (2022).
19. Bal, R., Xiao, Y. & Wang, W. PGraphDTA: Improving drug–target interaction prediction using protein language models and contact maps (2023). 2310.04017.
20. Li, J. & Yao, L. HCAF-DTA: drug–target binding affinity prediction with cross-attention fused hypergraph neural networks (2025). 2504.02014.
21. Li, S. *et al.* MONN: a multi-objective neural network for predicting compound–protein interactions and affinities. *Cell Syst.* **10**, 308–322.e11, DOI: 10.1016/j.cels.2020.03.002 (2020).
22. Gim, M. *et al.* ArkDTA: attention regularization guided by non-covalent interactions for explainable drug–target binding affinity prediction. *Bioinformatics* **39**, i448–i457, DOI: 10.1093/bioinformatics/btad207 (2023).

23. Seo, S., Kim, H., Lee, J., Choi, S. & Park, S. Exploring the potential of compound–protein complex structure-free models in virtual screening using BlendNet. *Briefings Bioinforma.* **26**, bbae712, DOI: 10.1093/bib/bbae712 (2024).

24. Wen, M. *et al.* Deep-learning-based drug–target interaction prediction. *J. proteome research* **16**, 1401–1409, DOI: 10.1021/acs.jproteome.6b00618 (2017).

25. Öztürk, H., Ozkirimli, E. & Özgür, A. WideDTA: prediction of drug-target binding affinity (2019). 1902.04166.

26. Chen, L. *et al.* TransformerCPI: improving compound–protein interaction prediction by sequence-based deep learning with self-attention mechanism and label reversal experiments. *Bioinformatics* **36**, 4406–4414, DOI: 10.1093/bioinformatics/btaa524 (2020).

27. Zhao, Q., Xiao, F., Yang, M., Li, Y. & Wang, J. AttentionDTA: prediction of drug–target binding affinity using attention model. In *2019 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 64–69, DOI: 10.1109/BIBM47256.2019.8983125 (IEEE, 2019).

28. Rives, A. *et al.* Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proc. Natl. Acad. Sci.* **118**, e2016239118, DOI: 10.1073/pnas.2016239118 (2021).

29. Voitsitskyi, T. *et al.* 3DProtDTA: a deep learning model for drug–target affinity prediction based on residue-level protein graphs. *RSC Adv.* **13**, 10261–10272, DOI: 10.1039/d3ra00281k (2023).

30. Liu, S. *et al.* Versatile framework for drug–target interaction prediction by considering domain-specific features. *J. Chem. Inf. Model.* **64**, 5646–5656, DOI: 10.1021/acs.jcim.4c00403 (2024).

31. Kalemati, M., Zamani Emani, M. & Koohi, S. InceptionDTA: Predicting drug–target binding affinity with biological context features and inception networks. *Heliyon* **11**, e42476, DOI: 10.1016/j.heliyon.2025.e42476 (2025).

32. Jiménez-Luna, J., Grisoni, F. & Schneider, G. Drug discovery with explainable artificial intelligence. *Nat. Mach. Intell.* **2**, 573–584, DOI: 10.1038/s42256-020-00236-4 (2020).

33. Huang, K. *et al.* Therapeutics data commons: Machine learning datasets and tasks for drug discovery and development (2021). 2102.09548.

34. Hu, W. *et al.* Strategies for pre-training graph neural networks (2020). 1905.12265.

35. Kyte, J. & Doolittle, R. F. A simple method for displaying the hydropathic character of a protein. *J. Mol. Biol.* **157**, 105–132, DOI: 10.1016/0022-2836(82)90515-0 (1982).

36. Selvaraju, R. R. *et al.* Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *2017 IEEE International Conference on Computer Vision (ICCV)*, 618–626, DOI: 10.1109/ICCV.2017.74 (2017).

37. Gilson, M. K. *et al.* Bindingdb in 2015: A public database for medicinal chemistry, computational chemistry and systems pharmacology. *Nucleic Acids Res.* **44**, D1045–D1053, DOI: 10.1093/nar/gkv1072 (2016).

38. Fey, M. & Lenssen, J. E. Fast graph representation learning with PyTorch Geometric (2019). 1903.02428.

39. RDKit Community. RDKit: Open-source cheminformatics. https://www.rdkit.org/ (2023).

40. Zhang, J., Yang, P. L. & Gray, N. S. Targeting cancer with small molecule kinase inhibitors. *Nat. Rev. Cancer* **9**, 28–39, DOI: 10.1038/nrc2559 (2009).

41. Salentin, S., Schreiber, S., Haupt, V. J., Adasme, M. F. & Schroeder, M. Plip: fully automated protein–ligand interaction profiler. *Nucleic acids research* **43**, W443–W447, DOI: 10.1093/nar/gkv315 (2015).

42. Wu, N., Strömich, L. & Yaliraki, S. N. Prediction of allosteric sites and signaling: Insights from benchmarking datasets. *Patterns* **3**, 100408, DOI: 10.1016/j.patter.2021.100408 (2022).

43. Gatlin, W. *et al.* Decoding allosteric grammar with explainable AI integrating protein language models and energy landscape analysis: Neutral frustration at allosteric binding sites encodes regulatory versatility in protein kinases, DOI: 10.64898/2026.03.21.713412 (2026).

44. Zhang, S. & Bowman, G. R. Decrypting cryptic pockets with physics-based simulations and artificial intelligence. *Curr. Opin. Struct. Biol.* **96**, 103215, DOI: 10.1016/j.sbi.2025.103215 (2026).

45. Škrhák, V., Novotný, M., Feidakis, C. P., Krivák, R. & Hoksza, D. CryptoBench: cryptic protein–ligand binding sites dataset and benchmark. *Bioinformatics* **41**, btae745, DOI: 10.1093/bioinformatics/btae745 (2024).